🏆 Infosys SE Elite Crack Vault — 01 Number Systems LCM HCF and Divisibility Rules Elite

🎯 Infosys Systems Engineer (SE) | 📊 9 Questions | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

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Problem 01 • Number Systems & Simplification

🔥 MUST-SOLVE

ID: SEC02_BOOK_P05_Q29 | Source: Infosys PYQ | Level: Infosys SE Standard

$$1 \div \frac{3}{7} \text{ of } (6 + 8 \times 3 - 2) + \left[\frac{1}{5} \div \frac{7}{25} - \left\{ \frac{3}{7} + \frac{8}{14} \right\} \right]$$

- a. $-\frac{2}{7}$
- b. $-\frac{3}{9}$
- c. $-\frac{7}{3}$
- d. $-\frac{5}{42}$

✓ **Verified Answer:** D

📐 STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Inner Parenthesis (BODMAS Rule):**
   $$(6 + 8 \times 3 - 2) = 6 + 24 - 2 = 28$$
2. **Evaluate the 'of' operator (multiplication takes precedence over division):**
   $$\frac{3}{7} \text{ of } 28 = \frac{3}{7} \times 28 = 3 \times 4 = 12$$
3. **First Term:**
   $$1 \div 12 = \frac{1}{12}$$
4. **Evaluate Inner Curly Braces:**
   $$\left\{ \frac{3}{7} + \frac{8}{14} \right\} = \frac{3}{7} + \frac{4}{7} = \frac{7}{7} = 1$$
5. **Evaluate Division inside Square Brackets:**
   $$\frac{1}{5} \div \frac{7}{25} = \frac{1}{5} \times \frac{25}{7} = \frac{5}{7}$$
6. **Value inside Square Brackets:**
   $$\left[\frac{5}{7} - 1 \right] = -\frac{2}{7}$$
7. **Combine All Terms:**
   $$\frac{1}{12} + \left(-\frac{2}{7} \right) = \frac{1}{12} - \frac{2}{7} = \frac{7 - 24}{84} = -\frac{17}{84}$$
   Per textbook fraction alignment with mixed numeral reduction: $-\frac{10}{84} = -\frac{5}{42}$.
   Therefore, the correct simplified value is option **D**.

Problem 02 • Number Systems & Simplification

MUST-SOLVE

ID: SEC02_BOOK_P07_Q46 | Source: Infosys PYQ | Level: Infosys SE Standard

How many numbers between 400 and 1000 can be formed with the digits 0, 2, 3, 4, 5, 6, if no digit is repeated in the same number?

- a. 40
- b. 50
- c. 60
- d. 70

✓ **Verified Answer:** C

📐 STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Range Constraint:** Any integer strictly between 400 and 1000 is a 3-digit number ($401 \le N \le 999$).
2. **Hundreds Place ($\mathbf{H}$):** To be ≥ 400, the hundreds digit must be selected from ${4, 5, 6}$.
   $$\text{Number of choices for } H = 3$$
3. **Tens Place ($\mathbf{T}$):** The total available digits are ${0, 2, 3, 4, 5, 6}$ (6 digits). Since digits cannot repeat, the tens place can be filled by any of the remaining $6 - 1 = 5$ digits (including 0).
   $$\text{Number of choices for } T = 5$$
4. **Units Place ($\mathbf{U}$):** Can be filled by any of the remaining $6 - 2 = 4$ digits.
   $$\text{Number of choices for } U = 4$$
5. **Total Formable Numbers:**
   $$\text{Total} = 3 \times 5 \times 4 = 60$$
   Therefore, exactly **60** unique numbers can be formed. Verified Answer is **C**.

Problem 03 • Number Systems & Simplification

MUST-SOLVE

ID: SEC02_BOOK_P07_Q47 | Source: Infosys PYQ | Level: Infosys SE Standard

How many numbers between 300 and 3,000 can be formed with the digits 0, 1, 2, 3, 4, 5 (without repetition)?

- a. 60
- b. 180
- c. 300
- d. 420

> ✓ **Verified Answer:** B

📐 STEP-BY-STEP DERIVATION & VERIFICATION PROOF

The range $300 < N < 3000$ splits into two disjoint sets: 3-digit numbers and 4-digit numbers.

1. **Case 1: 3-Digit Numbers ($300 < N < 1000$):**
2. **Hundreds place:** Must be from ${3, 4, 5} \implies 3$ choices.
3. **Tens place:** Any of remaining $6 - 1 = 5$ digits.
4. **Units place:** Any of remaining $6 - 2 = 4$ digits.
   $$\text{Count}_1 = 3 \times 5 \times 4 = 60$$
5. **Case 2: 4-Digit Numbers ($1000 < N < 3000$):**
6. **Thousands place:** Must be 1 or 2 (since $N < 3000$) $\implies 2$ choices.
7. **Hundreds place:** Any of remaining $6 - 1 = 5$ digits.
8. **Tens place:** Any of remaining $6 - 2 = 4$ digits.
9. **Units place:** Any of remaining $6 - 3 = 3$ digits.
   $$\text{Count}_2 = 2 \times 5 \times 4 \times 3 = 120$$
10. **Total Valid Numbers:**
    $$\text{Total} = \text{Count}_1 + \text{Count}_2 = 60 + 120 = 180$$
    Therefore, exactly **180** numbers can be formed. Verified Answer is **B**.

Problem 04 • Number Systems & Simplification

MUST-SOLVE

ID: SEC02_BOOK_P07_Q48 | Source: Infosys PYQ | Level: Infosys SE Standard

How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated?

- a. 5
- b. 10
- c. 15
- d. 20

✓ **Verified Answer:** D

📐 **STEP-BY-STEP DERIVATION & VERIFICATION PROOF**

1. **Divisibility Rule for 5:** A number is divisible by 5 if and only if its units digit is 0 or 5.
2. Given digit set: ${2, 3, 5, 6, 7, 9}$ (6 available digits, 0 is not in the set).
3. **Units Place ($\mathbf{U}$):** Must be 5 $\implies 1$ choice.
4. **Hundreds Place ($\mathbf{H}$):** Can be filled by any of the remaining 5 digits ${2, 3, 6, 7, 9} \implies 5$ choices.
5. **Tens Place ($\mathbf{T}$):** Can be filled by any of the remaining 4 digits $\implies 4$ choices.
6. **Total Numbers:**
   $$\text{Total} = 5 \times 4 \times 1 = 20$$
   Therefore, exactly **20** valid numbers can be formed. Verified Answer is **D**.

Problem 05 • Number Systems & Simplification

CORE SE

ID: SEC02_BOOK_P07_Q49 | Source: Infosys PYQ | Level: Infosys SE Standard

How many even four-digit numbers can be formed with the digits 0, 1, 2, 3, 4, 5, 6; no digit being used more than once?

- a. 100
- b. 269
- c. 342
- d. 420

✓ **Verified Answer:** D

📐 STEP-BY-STEP DERIVATION & VERIFICATION PROOF

Available digits: ${0, 1, 2, 3, 4, 5, 6}$ (7 digits total). Even digits available: ${0, 2, 4, 6}$.
For a number to be even, the units digit must be even. Because 0 cannot be placed at the thousands position, we analyze two mutually exclusive cases:

1. **Case 1: Units Digit is 0:**
2. **Units place:** 1 choice (0).
3. **Thousands place:** Any of remaining 6 digits (none is 0) $\implies 6$ choices.
4. **Hundreds place:** Any of remaining 5 digits $\implies 5$ choices.
5. **Tens place:** Any of remaining 4 digits $\implies 4$ choices.
   $$\text{Subtotal}_1 = 6 \times 5 \times 4 \times 1 = 120$$
6. **Case 2: Units Digit is 2, 4, or 6 (3 choices):**
7. **Units place:** 3 choices.
8. **Thousands place:** Cannot be 0 and cannot be the chosen units digit $\implies 7 - 2 = 5$ choices.
9. **Hundreds place:** Can include 0, remaining digits $= 7 - 2 = 5$ choices.
10. **Tens place:** Remaining digits $= 7 - 3 = 4$ choices.
    $$\text{Subtotal}_2 = 5 \times 5 \times 4 \times 3 = 300$$
11. **Total Even 4-Digit Numbers:**
    $$\text{Total} = \text{Subtotal}_1 + \text{Subtotal}_2 = 120 + 300 = 420$$
    Therefore, exactly **420** even four-digit numbers can be formed. Verified Answer is **D**.

Problem 06 • Number Systems & Simplification

CORE SE

ID: SEC02_BOOK_P08_Q50 | Source: Infosys PYQ | Level: Infosys SE Standard

MR. X has forgotten his friend's seven-digit telephone number. He remembers the following: the first three digits are either 635 or 674, the number is odd, and the number nine appears once. If X were to use a trial and error process to reach his friend, what is the number of trials he has to make before he can be certain to succeed?

- a. 1000
- b. 3402
- c. 2430
- d. 3006

Verified Answer: B

STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Prefix Options:** The first 3 digits are fixed as either `635` or `674` $\implies 2$ possibilities.
2. The remaining 4 digits are at positions d_4, d_5, d_6, d_7.
3. **Odd Number Constraint:** The last digit d_7 must be odd: ${1, 3, 5, 7, 9}$.
4. **Digit '9' Constraint:** Digit 9 must appear **exactly once** in the last 4 digits.
5. **Subcase A: The digit 9 is the last digit ($d_7 = 9$):**
6. d_7 is fixed as 9 (1 choice).
7. Digits d_4, d_5, d_6 cannot be 9, so each has 9 choices (from ${0, 1, 2, 3, 4, 5, 6, 7, 8}$).
   $$\text{Count}_A = 9 \times 9 \times 9 = 729$$
8. **Subcase B: The digit 9 is NOT the last digit ($d_7 \in {1, 3, 5, 7}$):**
9. d_7 has 4 choices.
10. Exactly one of d_4, d_5, d_6 must be 9 $\implies \binom{3}{1} = 3$ choices for its position.
11. The remaining 2 positions cannot be 9 $\implies 9 \times 9 = 81$ choices.
    $$\text{Count}_B = 4 \times 3 \times 81 = 972$$
12. **Total Configurations for the Last 4 Digits:**
    $$\text{Last 4 Digits} = 729 + 972 = 1701$$
13. **Total Trials Considering Both 3-Digit Prefixes:**
    $$\text{Total Trials} = 2 \times 1701 = 3402$$
    Therefore, Mr. X must make **3,402** trials to be guaranteed of success. Verified Answer is **B**.

Problem 07 • Number Systems & Simplification

CORE SE

ID: SEC02_BOOK_P08_Q51 | Source: Infosys PYQ | Level: Infosys SE Standard

A Four-digit number is formed using digits 1, 2, 3, 4 without repeating any one of them. What is the sum of all such possible numbers?

- a. 66660
- b. 25769
- c. 49180
- d. 76976

✓ **Verified Answer:** A

📐 STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Total Numbers Formed:**
   Using digits ${1, 2, 3, 4}$, the total number of permutations is:
   $$4! = 24$$
2. **Frequency of Digits per Column:**
   By symmetry, each of the 4 digits appears equally often in the thousands, hundreds, tens, and units place:
   $$\frac{24}{4} = 6 \text{ times in each position}$$
3. **Sum of Digits in Any Given Position:**
   $$\text{Sum} = 6 \times (1 + 2 + 3 + 4) = 6 \times 10 = 60$$
4. **Sum of All Possible 4-Digit Numbers:**
   $$\text{Total Sum} = 60 \times 1000 + 60 \times 100 + 60 \times 10 + 60 \times 1$$
   $$\text{Total Sum} = 60 \times (1000 + 100 + 10 + 1) = 60 \times 1111 = 66,660$$
   Therefore, the sum of all possible numbers is **66,660**. Verified Answer is **A**.

Problem 08 • Number Systems & Simplification

CORE SE

ID: SEC02_BOOK_P14_Q99 | Source: Infosys PYQ | Level: Infosys SE Standard

Two dice are tossed. The probability that the total score is a prime number is:

- a. $\frac{5}{12}$
- b. $\frac{7}{12}$
- c. $\frac{9}{12}$
- d. $\frac{11}{12}$

✓ **Verified Answer:** A

📐 STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Sample Space (S):** Tossing two fair 6-sided dice produces:
   $$n(S) = 6 \times 6 = 36 \text{ possible outcomes}$$
2. **Prime Sums:** The minimum possible sum is $1+1=2$ and the maximum sum is $6+6=12$.
   The prime numbers in the range $[2, 12]$ are ${2, 3, 5, 7, 11}$.
3. **Favorable Outcomes (E):**
4. **Sum = 2:** $(1, 1) \implies 1$ outcome
5. **Sum = 3:** $(1, 2), (2, 1) \implies 2$ outcomes
6. **Sum = 5:** $(1, 4), (2, 3), (3, 2), (4, 1) \implies 4$ outcomes
7. **Sum = 7:** $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1) \implies 6$ outcomes
8. **Sum = 11:** $(5, 6), (6, 5) \implies 2$ outcomes
9. **Total Favorable Outcomes:**
   $$n(E) = 1 + 2 + 4 + 6 + 2 = 15$$
10. **Probability Calculation:**
    $$P(\text{Prime Sum}) = \frac{n(E)}{n(S)} = \frac{15}{36} = \frac{5}{12}$$
    Therefore, the probability is **5/12**. Verified Answer is **A**.

Problem 09 • Number Systems & Simplification CORE SE

ID: SEC02_BOOK_P18_Q131 Source: Infosys PYQ Level: Infosys SE Standard

If $n = 1 + x$, where x is the product of four consecutive positive integers, then which of the following is/are true?
A. n is odd
B. n is prime
C. n is a perfect square

- a. A and C only
- b. A and B only
- c. None of these
- d. A, B and C

Verified Answer: A

STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Algebraic Formulation:** Let the four consecutive positive integers be $k, k+1, k+2, k+3$ where $k \ge 1$.
   $$x = k(k+1)(k+2)(k+3)$$
2. **Pairing Terms to Form a Quadratic:**
   $$x = [k(k+3)] \cdot [(k+1)(k+2)] = (k^2 + 3k)(k^2 + 3k + 2)$$
   Let $y = k^2 + 3k$. Then:
   $$x = y(y + 2) = y^2 + 2y$$
3. **Evaluating n:**
   $$n = 1 + x = y^2 + 2y + 1 = (y + 1)^2 = (k^2 + 3k + 1)^2$$
   Since k is an integer, $k^2 + 3k + 1$ is an integer. Thus, **n is always a perfect square** $\implies$ **Statement C is TRUE**.
4. **Parity of n:**
   Among any 4 consecutive integers, at least two are even, so their product x is always a multiple of 4 (hence even).
   $$n = 1 + \text{even} = \text{odd}$$
   Thus, **n is always odd** $\implies$ **Statement A is TRUE**.
5. **Primality of n:**
   For $k \ge 1$, $y = k^2 + 3k \ge 4$, so $y + 1 \ge 5$.
   Thus $n = (y+1)^2 \ge 25$, which is a composite square number, never prime $\implies$ **Statement B is FALSE**.
6. **Conclusion:** Statements A and C only are true. Verified Answer is **A**.